

Diff 4**Review**

1. (b) $y = -x^2 - 2x + 8$

2.

$$p \propto \frac{1}{q} \rightarrow p = \frac{k}{q}$$

$$4 = \frac{k}{-2.5}$$

$$k = -1.6$$

Then

$$p = -\frac{1.6}{q}$$

$$q = -\frac{1.6}{0.5}$$

$$= -3.2$$

3. $y' = 4x - 1$. At $x = 1$,

$$y' = \nabla = 3$$

$$\nabla_{\text{normal}} = -\frac{1}{3}$$

$$\therefore y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{1}{3}x + \frac{1}{3}$$

$$y = -\frac{1}{3}x + \frac{13}{3}$$

Then the points where they intersect can be found by

$$2x^2 - x + 3 = -\frac{1}{3}x + \frac{13}{3}$$

$$6x - 2x - 4 = 0$$

$$3x - x - 2 = 0$$

$$(3x + 2)(x - 1) = 0$$

So the other x-coordinate (the x-coordinate of Q) is $-\frac{2}{3}$.

4. The area of a sector can be found by $\frac{1}{2}r^2\theta$, and the arc length of a sector can be found by $r\theta$. Then

$$8\theta = 3\pi$$

$$\theta = \frac{3}{8}\pi$$

So the area of the sector is

$$\frac{1}{2} \times 8^2 \times \frac{3}{8}\pi$$

$$= 12\pi$$

5. $4x - x^2 = x(4 - x)$, so the curve intersects the x-axis at $x = 0$ and $x = 4$. Then

$$\begin{aligned}\int_0^4 (4x - x^2) dx &= \left[2x^2 - \frac{1}{3}x^3 \right]_0^4 \\ &= \frac{32}{3}\end{aligned}$$

Consolidation

1. The curve is concave if y'' is less than 0.

i. $y'' = 6x + 4$. At $x = 1$, $y'' = 10 > 0$ so the curve is convex.

ii. $y'' = 12x - 6$. At $x = 0$, $y'' = -6 < 0$ so the curve is concave.

iii. $y'' = 12x^2 + 6$. At $x = -2$, $y'' = 54 > 0$ so the curve is convex.

2. For the curve $y = x^3 + 3x^2 + 3x + 4$, $y' = 3x^2 + 6x + 3$ and $y'' = 6x + 6$.

Then at stationary points, $y' = 0 \rightarrow 3(x + 1) = 0$. So there is a stationary point at $x = -1$.

At stationary points, if $y'' = 0$ the point is a stationary point of inflection. $y''(-1) = -6x + 6 = 0$, so the stationary point at $x = -1$ is a stationary point of inflection.

At $x = -1$, $y = 3$, so the coordinates of the stationary point are $(-1, 3)$.

3. i. $y' = 4x^3 - 6x^2$; $y'' = 12x^2 - 12x$. At stationary points, $y' = 0$

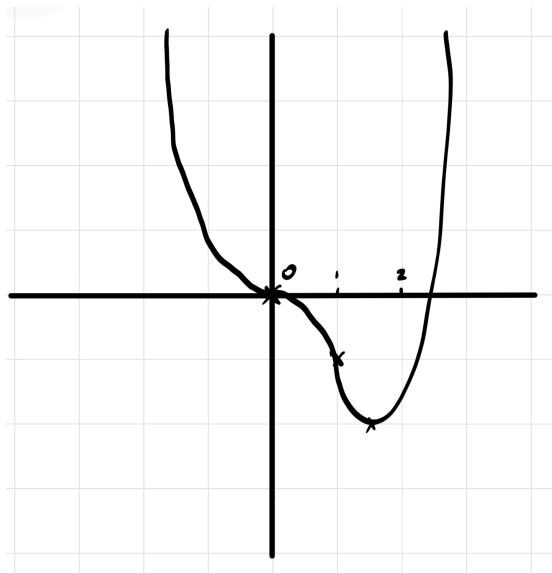
$$\rightarrow x^2(4x - 6) = 0 \text{ so } x = 0 \text{ or } x = \frac{3}{2}.$$

At $x = 0$, $y'' = 0$, so by using $y'(-1) = -10$ and $y'(1) = -2$, we get that the point on the graph where $x = 0$ is a stationary point of inflection.

At $x = \frac{3}{2}$, $y'' = 9$, so the point on the graph where $x = \frac{3}{2}$ is a minimum.

ii. The non-stationary point of inflection's x coordinate is the other solution of $x(12x - 12) \rightarrow x = 0$ or $x = 1$. So the coordinate is $(1, y(1)) = (1, -1)$.

iii.



4. i. $y' = 12x^3 - 48x^2 + 60x - 24 = 12(x^3 - 4x^2 + 5x - 2)$.

ii. $y' = 12(x^3 - 4x^2 + 5x - 2)$. $y'(2) = 0 \rightarrow (x - 2)$ is a factor of y' . So

$$\begin{aligned} y' &= 12(x - 2)(x^2 - 2x + 1) \\ &= 12(x - 2)(x - 1)^2 \end{aligned}$$

Since there are only two solutions here, $x = 2$ and $x = 1$, there are only two points where $\frac{dy}{dx} = 0$.

iii. For $x = 2$:

$$y'(1.9) = -0.972$$

$$y'(2.1) = 1.452$$

So the curve has a minimum at $x = 2$.

For $x = 1$:

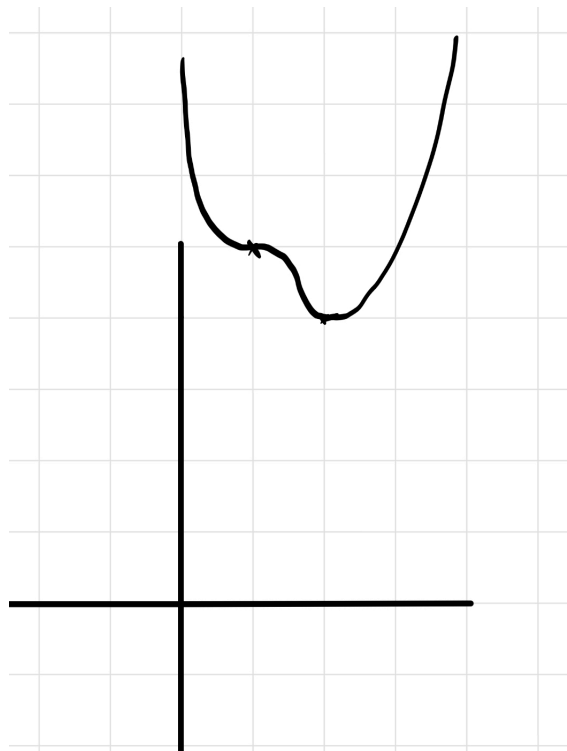
$$y'(0.9) = -0.132$$

$$y'(1.1) = -0.108$$

So the curve has a point of inflection at $x = 1$.

iv. At $x = 2$, $y = 4$. At $x = 1$, $y = 5$.

v.



5. i. $y = (x + 1)(x^3 - 9x^2 + 27x - 27)$

$$= x^4 - 8x^3 + 18x^2 + 0x - 27$$

ii. $y' = 4x^3 - 24x^2 + 36x$

When $y' = 0$,

$$4x(x^2 - 6x + 9) = 0$$

$$4x(x - 3)^2 = 0$$

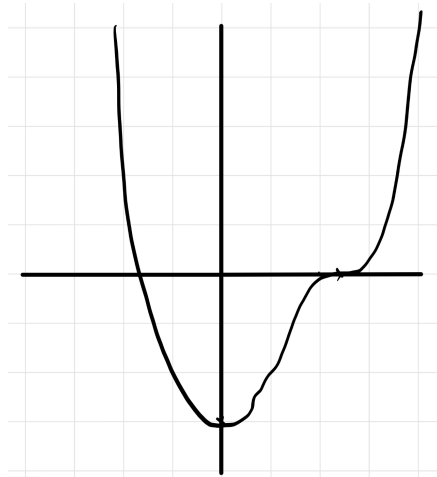
So $x = 0$ and $y = -27$ or $x = 3$ and $y = 0$.

iii. $y'' = 12x^2 - 48x + 36$.

At $x = 0$, $y'' = 36$ so the stationary point at $x = 0$ is a minimum.

At $x = 3$, $y'' = 0$ so the stationary point at $x = 3$ is a point of inflection.

iv.



6. i.

$$y' = -x^{-2} - 2x + 3$$

$$y'' = 2x^{-3} - 2$$

At stationary points, $y' = 0$, so

$$-\frac{1}{x^2} - 2x + 3 = 0$$

$$1 + 2x^3 - 3x^2 = 0$$

So stationary points at $x = -\frac{1}{2}$ and $x = 1$.

At $x = -\frac{1}{2}$, $y'' = -18$ so point is a maximum. At $x = 1$, $y'' = 0$ so point is a stationary point of inflection.

ii. The second derivative only has one real solution when $y'' = 0$, so there are no other points of inflection on the curve.

Exam Questions

1. $f(x) = \cos x$

$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\cos(x)(1 - \frac{1}{2}h^2) - h\sin(x) - \cos(x)}{h} \quad \text{since } h \text{ is small} \\
 &= \lim_{h \rightarrow 0} \frac{\cos x - \frac{1}{2}h^2 \cos x - h\sin(x) - \cos(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left(-\frac{1}{2}h \cos x - \sin(x) \right) \\
 &= -\sin x
 \end{aligned}$$

2. $y = x^2 + kx - 4x^{-1}$
 $y' = 2x + k + 4x^{-2}$

When $x = 2$, $y' = 0$. So

$$\begin{aligned}
 2(2) + k + \frac{4}{2^2} &= 0 \\
 k &= -5
 \end{aligned}$$

Then

$$y'' = 2 - \frac{8}{x^3}$$

When $y'' = 0$, the curve has a point of inflection .

$$\begin{aligned}
 \frac{8}{x^3} &= 2 \\
 x^3 &= 4
 \end{aligned}$$

So $x = \sqrt[3]{4}$, which is not a stationary point because $y'(\sqrt[3]{4}) \neq 0$.